Topic 2: AWS Site-to-Site VPN (Deeper Dive)

Complete Setup Steps

1. **Create Customer Gateway:** Register your on-premises public IP, select routing (Static or BGP), and provide your ASN (if using dynamic).
2. **Create Virtual Private Gateway:** Attach this to your VPC. Use the default AWS ASN or specify your own.
3. **Create VPN Connection:** Link the VGW and CGW. Set match routing (Static requires CIDRs, BGP requires ASN).
4. **Download Configuration:** Select your device vendor (Cisco, Juniper, etc.) to get the specific config file.
5. **Enable Route Propagation:** In your VPC Route Table, toggle propagation to allow VGW routes to propagate automatically.
6. **Update Security Groups:** Ensure on-premise IP CIDRs are permitted in your AWS security groups.

VPN Tunnel Architecture

Topology:

Your Device (203.0.113.1) ═════ Tunnel 1 ═════► AWS Endpoint 1

Your Device (203.0.113.1) ═════ Tunnel 2 ═════► AWS Endpoint 2

Key Features:

- Two tunnels generated automatically.
- Use BGP for automatic failover.
- Encryption standard: AES-256-GCM.

Important Concepts

- **Dead Peer Detection (DPD):** Automatically detects tunnel failure to trigger failover.
- **BGP ASN:** Your on-prem device needs a private ASN (64512-65534). Always ensure your side and AWS side use **different** ASNs.
- **Static vs. Dynamic:** Dynamic (BGP) is preferred for production to avoid manual route table management.

Troubleshooting Checklist

Checklist: Phase 1 IKE handshake mismatch? Check pre-shared keys. Check MTU/MSS clamping if large packets are failing but small pings succeed.